

Product Requirements Document: B2B SaaS Carbon Compliance Platform

1. Introduction

1.1. Purpose of the Document

This Product Requirements Document (PRD) outlines the functionality, features, and user experience for a B2B SaaS platform designed to assist Original Equipment Manufacturers (OEMs) and Small and Medium-sized Enterprises (SMEs) in achieving Scope 3 carbon compliance. It serves as a guiding blueprint for the development team, ensuring alignment across all stakeholders regarding the product's vision, scope, and implementation details.

1.2. Product Overview

The platform operates as a dual-purpose solution, encompassing a public-facing marketing site and a secure client dashboard. The marketing site is engineered to explain the platform's services, focusing on converting visitors into booked calls. The client dashboard, conversely, functions as the core product, enabling paying customers to upload supplier documents, receive precise Scope 3 carbon calculations, and generate audit-ready compliance reports. This architectural separation, akin to established platforms like Linear, allows for distinct user journeys while maintaining a unified brand presence under a single domain.

1.3. Goals

The primary goals of this platform are to:

- Facilitate compliance with stringent environmental regulations, particularly CSRD Wave 2 for European OEMs.
- Provide accurate and auditable Scope 3 Category 1 emissions data, addressing the challenges posed by non-digital ASEAN supply chains.
- Offer ESG readiness audits, framework guidance, and greenwashing protection for Singaporean SMEs.
- Streamline the process of supplier document management and carbon reporting.
- Enhance user experience through intuitive interfaces and intelligent AI assistance.

1.4. Target Audience

The platform caters to two distinct buyer personas:

Audience Type	Primary Focus	Key Needs	Compliance Drivers
Primary Buyer: Sustainability and Procurement Teams at Multinational European OEMs	Companies with ASEAN supply chains	Scope 3 Category 1 supplier emissions data; solutions for non-digital suppliers	CSRD Wave 2 compliance deadlines (2025-2026)
Secondary Buyer: Singapore SMEs	Local businesses	ESG readiness audits; framework guidance; greenwashing protection	Satisfy buyers and sustainability-linked lenders

2. Scope 3 Carbon Compliance Context

2.1. CSRD Wave 2 Compliance

The Corporate Sustainability Reporting Directive (CSRD) mandates comprehensive sustainability reporting for a broad range of companies within the European Union. Wave 2 of CSRD compliance, with reporting obligations commencing for financial years starting on or after January 1, 2025, significantly impacts large non-listed companies, including many multinational OEMs ¹ ². These entities are required to report on their environmental, social, and governance (ESG) impacts, with a particular emphasis on their value chain emissions.

2.2. Scope 3 Category 1 Emissions

Scope 3 emissions encompass all indirect greenhouse gas emissions occurring in a company's value chain, both upstream and downstream. Category 1, specifically, pertains to emissions from purchased goods and services ³. For OEMs, accurately calculating these emissions is critical but often challenging, especially when dealing with suppliers in regions like ASEAN where digital infrastructure for data collection may be limited. The platform aims to bridge this data gap by enabling the ingestion and processing of non-digital supplier documents.

2.3. Challenges with ASEAN Supply Chains

ASEAN (Association of Southeast Asian Nations) supply chains present unique challenges for Scope 3 data collection due to varying levels of digital maturity among suppliers. Many suppliers in this region may not have sophisticated systems for tracking and reporting their emissions data, making traditional data collection methods inefficient or impossible. The

platform's ability to process diverse document types and leverage AI for data extraction is crucial for overcoming these hurdles.

3. Product Features

This section details the current state and future development of the platform's features, categorized by their development status.

3.1. Existing Features

These features are currently operational and form the foundational elements of the platform:

- **Marketing Pages:** Audit, frameworks, greenwashing protection, and pricing pages are in place.
- **Client Dashboard UI:** Provides an interface for ESG compliance scoring.
- **AI Chat Assistant (Aria):** Powered by Gemini, offering user support and insights.
- **User Authentication:** Login, registration, and password change functionalities are handled via Cloudflare Pages Functions and Supabase.
- **Contact/Book Call Form:** Submits data to Supabase for lead management.

3.2. Features in Progress (Frontend Shells)

These features have a frontend interface but require backend development to become fully functional:

- **Document Upload Flow:** Designed for supplier invoices and utility bills, awaiting backend integration for processing.
- **Scope 3 Carbon Calculation Engine:** Frontend is ready, but the backend integration with Layer 1 Claude API ingestion and a Layer 2 proxy model for calculations is pending.
- **Audit-Ready Report Generation & PDF Export:** Frontend components for report generation and export exist, but the underlying data processing and PDF creation are not yet implemented.

3.3. Planned Features

These features are slated for future development and will enhance the platform's core offerings:

- **ASEAN Regional Emission Factor Database:** This represents a core intellectual property (IP) and will provide region-specific emission factors for more accurate calculations.

- **Supplier Management:** Functionality to track multiple suppliers per OEM account, including their data and compliance status.
- **Confidence-Level Labelling:** Assigning confidence levels to all carbon outputs to enhance audit defensibility and transparency.
- **Methodology Transparency Page:** A dedicated section explaining the calculation methodologies for audit purposes.

3.4. Broken Features (Immediate Fixes)

These critical issues require immediate attention to ensure basic platform functionality and user experience:

- **openModal() Function Error:** The `openModal()` function currently throws a `ReferenceError` on every "Book Call" click, preventing users from initiating contact.
- **Mobile Hamburger Menu:** The mobile navigation menu disappears on mobile devices without a functional replacement, severely impacting mobile usability.
- **Missing Linked Pages:** Seven critical pages are currently missing: `audit.html`, `eu-audit.html`, `eu-frameworks.html`, `eu-protection.html`, `eu-cbam.html`, `gill.html`, and a functional contact page. These pages are essential for both marketing and compliance information.
- **Backend JS File Naming:** All backend JavaScript files are incorrectly named, which would lead to 404 errors for every API call in a production environment.

4. Technical Requirements

4.1. Architecture and Technologies

The platform leverages a modern web architecture with specific technologies for different functionalities:

- **Frontend:** Cloudflare Pages for hosting, likely React/Vue/Angular for UI (to be confirmed, but implied by modern SaaS).
- **Backend:** Cloudflare Pages Functions for serverless logic, Supabase for user authentication and database management.
- **AI/ML:** Gemini for the Aria chat assistant, Claude API for Layer 1 carbon calculation ingestion, and a custom Layer 2 proxy model for refined calculations.

4.2. Data Management

- **Supplier Document Ingestion:** The system must support the upload of various document types (e.g., PDFs, images of invoices, utility bills). OCR and intelligent document processing will be required to extract relevant data from non-digital formats.
- **Emission Factor Database:** A robust and scalable database is required to store and manage ASEAN regional emission factors, which is a core IP of the platform.
- **Audit Trail:** All carbon calculations and data modifications must be logged to provide a clear, auditable trail for compliance reporting.

4.3. Integrations

- **Supabase:** For user authentication and primary data storage.
- **Cloudflare Pages Functions:** For serverless backend logic and API endpoints.
- **Gemini API:** For the AI chat assistant, Aria.
- **Claude API:** For initial ingestion and processing of carbon-related data.

5. User Stories

This section outlines key user stories to illustrate the platform's functionality from the perspective of its target users.

5.1. OEM Sustainability Manager

- As an OEM Sustainability Manager, I want to upload supplier invoices and utility bills, so that I can get accurate Scope 3 Category 1 carbon emissions data for my ASEAN supply chain.
- As an OEM Sustainability Manager, I want to view ESG compliance scores for my suppliers, so that I can identify high-risk areas and drive improvements.
- As an OEM Sustainability Manager, I want to generate audit-ready compliance reports, so that I can meet CSRD Wave 2 reporting deadlines.
- As an OEM Sustainability Manager, I want to understand the methodology behind the carbon calculations, so that I can defend our reports during an audit.

5.2. Singapore SME Owner

- As a Singapore SME Owner, I want to access ESG readiness audits, so that I can understand my current sustainability performance.
- As a Singapore SME Owner, I want guidance on ESG frameworks, so that I can implement relevant sustainability practices.

- As a Singapore SME Owner, I want protection against greenwashing claims, so that I can maintain my company's reputation and satisfy lenders.

5.3. General User (Marketing Site Visitor)

- As a potential client, I want to easily find information about the platform's services, so that I can understand how it can help my business.
- As a potential client, I want to book a call with a sales representative, so that I can discuss my specific needs and get a demo.

6. Success Metrics

Key performance indicators (KPIs) to measure the success of the platform include:

- **Marketing Site Conversion Rate:** Percentage of visitors who book a call or submit a contact form.
- **Client Dashboard User Engagement:** Frequency and duration of client dashboard usage, particularly for document uploads and report generation.
- **Data Accuracy:** Percentage of Scope 3 Category 1 calculations that are deemed audit-ready and defensible.
- **Compliance Rate:** Number of clients successfully meeting CSRD Wave 2 or other relevant ESG reporting requirements.
- **Bug Resolution Rate:** Speed and effectiveness in addressing critical bugs like the `openModal()` error and mobile menu issues.
- **Feature Adoption:** Rate at which new features (e.g., supplier management, confidence-level labelling) are utilized by clients.

7. Future Considerations

This section outlines potential future enhancements and strategic directions for the platform.

- **Expansion to other Scope 3 Categories:** Investigate and implement calculations for other relevant Scope 3 categories beyond Category 1.
- **Integration with ERP/Procurement Systems:** Develop integrations with common OEM ERP and procurement systems (e.g., SAP, Ariba) to automate data flow.
- **Advanced Analytics and Benchmarking:** Provide clients with deeper insights into their emissions data, including benchmarking against industry peers.

- **Machine Learning for Anomaly Detection:** Implement ML models to identify anomalies in supplier data or carbon calculations, improving data quality and audit defensibility.
- **Multi-language Support:** Expand the platform to support multiple languages to better serve a global user base.

8. References

- [1] Corporate sustainability reporting - Finance - European Commission:
- [2] The Corporate Sustainability Reporting Directive (CSRD), explained - Normative:
- [3] Purchased goods and services emissions (Scope 3 Category 1) - Plana.earth: